JOHN ZAYAT

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PROFILE

MEng Aeronautical Engineering student graduating in 2027, with hands-on experience designing, manufacturing, integrating and testing uncrewed aircraft and electromechanical systems. Skilled in Siemens NX, first-principles analysis, FDM prototyping, hardware testing and failure investigation, progressing projects from requirements and CAD through manufacture, test and redesign.

ENGINEERING PROJECTS

ARCHANGEL – Independent Autonomous VTOL CCA Development | *Oct. 2023 – Present*

- Independently progressed a 1.7 m, 8 kg VTOL/CTOL aircraft through five major design iterations, manufacturing and assembling three physical airframes using 3D-printed structures, FDM prototyping and carbon-fibre reinforcement.
- Integrated five EDF motors, ESCs, power distribution, CubePilot flight control, servos and sensors; designed custom landing gear, EO/IR turret and thrust-vectoring/variable-nozzle mechanisms.
- Applied first-principles sizing, STAR-CCM+ CFD and Simcenter 3D structural analysis to evaluate aerodynamic performance, propulsion integration and airframe reinforcement.
- Planned and conducted six system-level hover tests, synchronising flight logs, video and audio in MATLAB to investigate thrust loss, asymmetric propulsion and pitch instability and define redesign requirements.
- Developed 6-DOF MATLAB/Simulink and Mission Planner simulations to evaluate VTOL stability, control response and recorded test behaviour.

Loughborough University UAV Team — Aerodynamics Engineer | *Sept. 2025 - Present*

- Batch-analysed multiple aerofoil profiles and wing dihedral/anhedral configurations in XFLR5, supporting aerodynamic selection for a 10kg, autonomous X-wing-inspired tandem-wing UAV.
- Performed aircraft stability calculations and XFLR5 simulations to evaluate longitudinal and directional stability, informing vertical-stabiliser and control-surface sizing.
- Conducted full-airframe STAR-CCM+ CFD simulations to assess EDF intake placement and wake interaction between the forward and rear wings.
- Designed internal EDF S-duct geometry in Siemens NX and evaluated internal-flow behaviour in STAR-CCM+ to support propulsion integration.

Two-Shaft Turbofan Engine Design - University Group Project | *Sept. 2025 – May 2026*

- Completed the preliminary mechanical and aerodynamic design of a two-spool turbofan engine, producing 13.34 kN thrust at Mach 0.82 and 35,000 ft for a business-aircraft application.
- Developed the thermodynamic cycle in GasTurb 12; used iterative hand calculations to size the engine's gas-path; designed the mechanical architecture and cross-sectional assembly layout.
- **Awarded the Stan Stevens Prize** for the best individual performance in the Gas Turbine Design module cohort.

PROFESSIONAL ENGINEERING WORK EXPERIENCE

Preventative Maintenance Engineer – *Marriott Hotels Twickenham* | *Sept. 2022 – Sept. 2023*

- Diagnosed and repaired HVAC, mechanical, electrical and plumbing systems across an operational hotel and sports stadium environment, while minimising disruption to guests.
- Developed an Excel-based tracking system that reduced maintenance turnaround time by 67%.
- Coordinated repairs with contractors and hotel departments, prioritising urgent faults, maintaining records and escalating specialist work where required.

EDUCATION

MEng Aeronautical Engineering - Loughborough University | *2023 – Present*

- Expected graduation date: 2027

A-Levels – Orleans Park School Sixth Form | *2021 - 2023*

- Obtained: Physics (A), Mathematics (A*), Further Mathematics (B) and EPQ (A*)
- EPQ: Investigated impact dynamics, composite materials, and crash safety technologies in Formula 1 & aerospace applications; achieved 100% and awarded with the **2023 EPQ Prize**.

TECHNICAL SKILLS

- **CAD, CFD & FEA Tools:** Siemens NX · SolidWorks · CATIA · STAR-CCM+ · XFLR5 · NX NASTRAN
- **Programming & Data Analysis:** MATLAB · Simulink · Python · Power BI · Tableau · SQL · Excel
- **Engineering & Maintenance Techniques:** Engineering Drawings · Root-Cause Analysis/Failure Investigation · Familiar with Technical Manuals/Publications · Hands-on Troubleshooting
- **Electronics & Manufacturing:** FDM Prototyping · 3D Printing · Carbon-Fibre Reinforcement · Mechanical Assembly · Wiring/Soldering · Motors/ESCs/Sensors/Servos/Flight Controllers · Electronics/Hardware Testing
- **Additional:** High Attention to Detail · Organization Skills · Persistence · Problem Solving · Team Communication and Coordination · Technical Report Writing

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CERTIFICATIONS

- **NX** Design Associate Certification, awarded by Siemens Digital Industries Software (2025)
- **CATIA** – Perform as a Mechanical Designer, awarded by Dassault Systèmes (2025)
- **SOLIDWORKS** 3D CAD for Education, awarded by Dassault Systèmes (2025)
- **MATLAB** Onramp Certifications in (awarded by MathWorks, 2025):
    - Simulink, SimScape, Control Design, Multibody Simulation, Signal Processing & Machine Learning

INTRODUCTORY COURSES

- Introduction to **Model-Based Systems Engineering (MBSE)**, awarded by Siemens (2025)
- Introduction to **Tableau**, awarded by Tableau (2025)
- **SQL** Foundations, awarded by Microsoft (2025)
- Data Analysis and Visualization with **Power BI**, awarded by Microsoft (2025)
- **Python** for Data Science, AI & Development, awarded by IBM (2025)

EXTRACURRICULAR ACTIVITIES

- Member of Loughborough University's Social Development Powerlifting Team and Drone Society
- Have flown a Piper PA28 Cherokee in pursuit of a Private Pilot License (PPL)
- Have a Flyer and Operator ID from the UK CAA for drone and model aircraft
- Student Affiliate of Royal Aeronautical Society (RAeS) and Institution of Mechanical Engineers (IMechE)